

JEE Advanced Booster Test - 4 | 2022

Date: 08/02/2021

Maximum Marks: 180

Timing: 04:00 PM - 09:00 PM

Duration: 3.0 Hrs

General Instructions

- The question paper consists of 3 Parts (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **four** sections (**Section 1, Section 2, Section 3 and Section 4**).
- Section 1** contains **6 Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

Section 2 contains **2 Paragraphs containing 2 Questions** on each paragraph. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

Section 3 contains **2 Column-Match sets**. Each set contains statements given in two columns (I and II). One or more statements in column II can be matched with a statement in column I. Each set has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

Section 4 contains **6 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
- For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code**, **Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

MARKING SCHEME

SECTION - 1

- This section consists of **SIX (06)** Questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+4 If only (all) the correct option(s) is(are) chosen
Partial Marks:	+3 If all the four options are correct but ONLY three options are chosen
Partial Marks:	+2 If three or more options are correct but ONLY two options are chosen and both of which are correct
Partial Marks:	+1 If two or more options are correct but ONLY one option is chosen, and it is a correct option
Zero Mark:	0 if none of the options is chosen (i.e. the question is unanswered)
Negative Marks:	-1 In all other cases.

SECTION - 2

- This section has **TWO (02) Paragraphs containing TWO (02) Questions** on each paragraph. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+3 If ONLY the correct option is chosen
Zero Marks:	0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks:	-1 In all other cases.

SECTION - 3

- This section has **TWO (02) Column-Match sets**. Each set contains statements given in two columns. Statements in column I have to be matched with statements in column II. One or more than one statements in column II can be matched with a statement in column I.
- Each set has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+3 If ONLY the option corresponding to the correct combination is chosen
Zero Marks:	0 If none of the options is chosen (i.e., the question is unanswered)
Negative Marks:	-1 In all other cases.

SECTION 4

- This section contains **6 Integer Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+3 If ONLY the correct Integer value is entered. There is NO negative marking .
Zero Marks:	0 In all other cases.

PART I : PHYSICS**60 MARKS****SECTION 1**

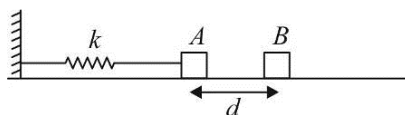
This section consists of SIX (06) Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

1. A large cylindrical container contains a non-viscous liquid of density 3ρ . A small cylinder of height L and density ρ is held vertically with its lower surface touching the liquid, as shown. The cylinder is now released, and it subsequently executes SHM. Which of these options is/are correct?
(Assume that the depth of liquid in the container is sufficient so that the cylinder does not touch the bottom)

- (A) Amplitude of the SHM is $\frac{2L}{3}$
 (B) Amplitude of the SHM is $\frac{L}{3}$
 (C) Time period of the SHM is $2\pi\sqrt{\frac{L}{3g}}$
 (D) Time period of the SHM is $2\pi\sqrt{\frac{2L}{3g}}$



2. Two spheres, A and B, made of the same metal and of radius R and $\frac{3R}{2}$ are heated to temperature $3T_0$ and $2T_0$ respectively, and suspended far away from each other in a vacuum chamber whose walls are maintained at temperature T_0 . Temperatures given are in kelvin. Assuming that the spheres are black bodies, and that they have high thermal conductivity so their entire volume is always at the same temperature, which of these options is/are correct?
- (A) Their initial rates of heat loss are in the ratio 128:81
 (B) Their initial rates of heat loss are in the ratio 64:27
 (C) Their initial rates of cooling are in the ratio 8:1
 (D) Their initial rates of cooling are in the ratio 9:2
3. Two small blocks, A and B, of mass m and $2m$, are placed on a smooth horizontal surface, and A is attached to a spring of spring constant k whose other end is fixed to a wall as shown. Initially, both blocks are at rest with a distance d between them. Now, the block A is moved a distance $2d$ to the left and released. The collision between the blocks is elastic. After the collision:



- (A) B moves at speed $\sqrt{\frac{4k}{3m}}d$
 (B) B moves at speed $\sqrt{\frac{k}{3m}}d$
 (C) A oscillates with amplitude $\frac{2d}{\sqrt{3}}$
 (D) A oscillates with amplitude $\frac{d}{\sqrt{3}}$

4. One mole of a monoatomic gas at initial pressure P_0 and initial temperature T_0 undergoes an adiabatic expansion. The work done by the gas in the process is $\frac{3RT_0}{4}$ (R is the universal gas constant). Which of the following statements is/are correct?
- (A) The average kinetic energy per molecule doubles
- (B) The final pressure of the gas is $\frac{P_0}{4\sqrt{2}}$
- (C) The final temperature of the gas is $\frac{T_0}{2}$
- (D) If the gas is now taken through an isobaric process back to the temperature T_0 , the amount of heat absorbed by the gas will be $\frac{5RT_0}{4}$
5. 1 mole of Argon gas (molar mass = 40 g/mol) at 460 K is mixed in a rigid, insulated container with 1 mole of Nitrogen gas (molar mass = 28 g/mol) at 300 K. After equilibrium is attained: (R is the gas constant)
- (A) The total kinetic energy of the molecules of the gas mixture is $1440R$
- (B) The temperature of the gas mixture is 360 K
- (C) The ratio of the root mean square speed of Argon molecules to the root mean square speed of Nitrogen molecules is $\sqrt{\frac{10}{7}}$
- (D) The ratio of the total kinetic energy of Argon molecules to the total kinetic energy of Nitrogen molecules is $\frac{3}{5}$
6. A cylindrical glass beaker of capacity V_0 and height H_0 at temperature T_0 is filled to the brim with a liquid also at temperature T_0 . The volume thermal expansion coefficient of the glass and the liquid is γ and 4γ respectively. The beaker is now heated to temperature $T_0 + \Delta T$, and some liquid spills out as a result. Then, the beaker is cooled to temperature $T_0 - \Delta T$, and the height of the liquid column inside the beaker becomes $H_0 - \Delta H$. Which of these options is/are correct?
- (A) The volume of the liquid that spilled out is $3V_0\gamma\Delta T$
- (B) The volume of the liquid that spilled out is $4V_0\gamma\Delta T$
- (C) ΔH is close to $\frac{13}{3}H_0\gamma\Delta T$
- (D) ΔH is close to $\frac{19}{3}H_0\gamma\Delta T$

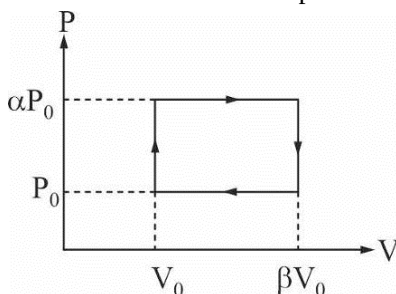
SPACE FOR ROUGH WORK

SECTION 2

This section has **TWO (02) Paragraphs** containing **TWO (02) Questions** on each paragraph. Each question has **FOUR options**. **ONLY ONE** of these four options is the correct answer.

Paragraph for Q.7 to Q.8

An ideal monoatomic gas at pressure P_0 and temperature T_0 occupies volume V_0 . The gas is now taken through a thermodynamic cycle comprising two isobaric and two isochoric processes, shown in the P-V diagram.



7. The maximum temperature of the gas during the cycle is:

(A) $\alpha(\beta - 1)T_0$	(B) $(\alpha - 1)\beta T_0$
(C) $(\alpha - 1)(\beta - 1)T_0$	(D) $\alpha\beta T_0$
8. The change in internal energy of the gas during the isochoric cooling is:

(A) $-\frac{3}{2}\alpha(\beta - 1)P_0V_0$	(B) $\frac{3}{2}\alpha(\beta - 1)P_0V_0$
(C) $-\frac{3}{2}(\alpha - 1)\beta P_0V_0$	(D) $\frac{3}{2}(\alpha - 1)\beta P_0V_0$

Paragraph for Q.9 to Q.10

Two rods P and Q , made of different materials, are placed end to end between two rigid walls separated by distance 3ℓ as shown. The rods P and Q respectively have area of cross-section A and A , length ℓ and 2ℓ , Young's modulus Y and $2Y$, and linear thermal expansion coefficient α and 2α . The rods are initially in thermal equilibrium with the surroundings and have negligible compressive stress in them. Now, the temperature of both rods is raised uniformly by ΔT .



9. The force now exerted by the wall on rod P is:

(A) $\frac{3}{2}AY\alpha\Delta T$	(B) $\frac{1}{2}AY\alpha\Delta T$	(C) $\frac{5}{2}AY\alpha\Delta T$	(D) $\frac{9}{2}AY\alpha\Delta T$
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10. The junction between the rods shifts a distance:

(A) $\frac{3}{2}\ell\alpha\Delta T$ rightwards	(B) $\frac{3}{2}\ell\alpha\Delta T$ leftwards
(C) $\frac{1}{2}\ell\alpha\Delta T$ rightwards	(D) $\frac{1}{2}\ell\alpha\Delta T$ leftwards

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SECTION 3

This section contains 2 Column-Match sets. Each set contains statements given in two columns (I and II). One or more statements in column II can be matched with a statement in column I. Each set has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

11. A particle executes SHM on the X-axis. Its position as a function of time is: $x(t) = A \sin\left(\omega t + \frac{\pi}{3}\right)$. Match each event in List-I to the correct time instant when this event **first occurs after** $t = 0$.

COLUMN-I		COLUMN-II	
(I)	The particle is at its mean position	(P)	$t = \frac{\pi}{6\omega}$
(II)	The particle has the same speed as at $t = 0$	(Q)	$t = \frac{\pi}{8\omega}$
(III)	The particle is at rest	(R)	$t = \frac{\pi}{3\omega}$
(IV)	The acceleration of the particle is half its maximum acceleration	(S)	$t = \frac{\pi}{4\omega}$
		(T)	$t = \frac{\pi}{2\omega}$
		(U)	$t = \frac{2\pi}{3\omega}$

Codes:

- (A) (I)-U; (II)-Q; (III)-P; (IV)-S
 (B) (I)-U; (II)-R; (III)-P; (IV)-T
 (C) (I)-Q; (II)-R; (III)-U; (IV)-T
 (D) (I)-Q; (II)-P; (III)-U; (IV)-R

12. Match the following columns.

Column I		Column II	
(I)	Isobaric heating	(P)	$\Delta Q > 0, \Delta U = 0$
(II)	Isothermal expansion	(Q)	$\Delta Q < 0, \Delta U = 0$
(III)	Adiabatic compression	(R)	$\Delta Q > 0, \Delta U > 0$
(IV)	Isochoric cooling	(S)	$\Delta Q = 0, \Delta U < 0$
		(T)	$\Delta Q = 0, \Delta U > 0$
		(U)	$\Delta Q < 0, \Delta U < 0$

Code:

- (A) I-P ; II-Q ; III-T ; IV-Q
 (B) I-R ; II-P ; III-S ; IV-U
 (C) I-R ; II-Q ; III-S ; IV-Q
 (D) I-R ; II-P ; III-T ; IV-U

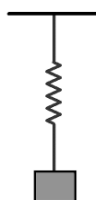
SPACE FOR ROUGH WORK

SECTION 4

This section has Six (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.

(Example: 6, 81, 1.50, 3.25, 0.08)

13. A wooden block of mass 12 kg is suspended in equilibrium from an ideal spring of spring constant 300 N/m as shown. A small bullet of mass 48 g is fired vertically into the block from below. The velocity of the bullet just before it strikes the block is 50 m/s. The bullet gets embedded in the block. The maximum height (to the nearest integer in cm) that the block-bullet system rises above its initial position is _____.



14. A calorimeter of water equivalent 50 g contains 150 g water at 20 °C. The amount of ice (in g) at –40 °C that must be added to the calorimeter, so that the final temperature is 0 °C with all the water frozen, is _____.

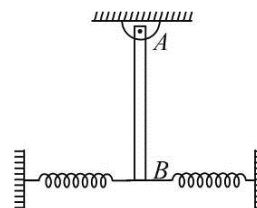
(Specific heat of water = 1 cal / g °C, specific heat of ice = 0.5 cal / g °C, latent heat of fusion of ice = 80 cal / g)

15. A gas with $\frac{C_p}{C_v} = \frac{3}{2}$ is heated at constant pressure. The ratio of the work done by the gas to the heat absorbed by the gas, $\frac{W}{\Delta Q}$ is equal to _____.

16. A hot body placed in a room maintained at a constant temperature T_0 °C takes the same amount of time to cool from 60 °C to 50 °C as it takes to cool from 40 °C to 35 °C. Assuming that the cooling process follows Newton's law of cooling, the value of T_0 (in °C) is _____.

17. A thin uniform rod of length ℓ , area of cross-section A , mass m , and Young's modulus Y is hinged at one of its ends and rotated in the horizontal plane at a constant angular velocity ω . The length of the rod increases by $X \left(\frac{m\omega^2 \ell^2}{AY} \right)$. The value of X is _____.

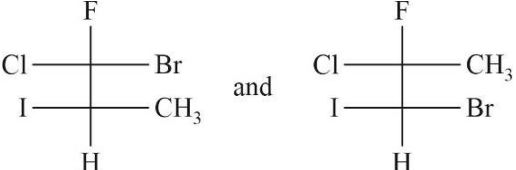
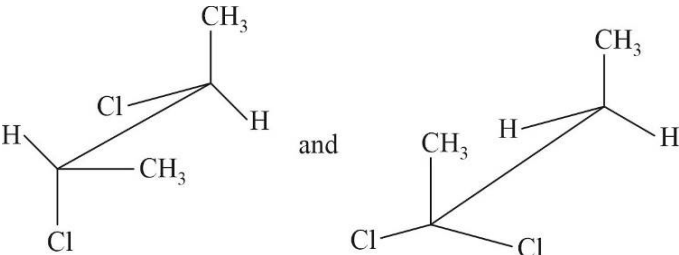
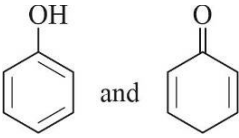
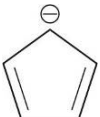
18. A thin, uniform rod of mass 3 kg and length 1 m is hinged at its top end A and connected to two ideal springs of spring constant 12.5 N/m each at its bottom end B. Initially both springs are in their natural length and the rod is in the vertical plane. If the rod is rotated slightly in the vertical plane and released, it begins oscillations of time period (in seconds) _____.
(Take $g = 10 \text{ m/s}^2$, $\pi = \sqrt{10}$)



SPACE FOR ROUGH WORK

PART II : CHEMISTRY**60 MARKS****SECTION 1**

This section consists of SIX (06) Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

- Which of the following solution have pH greater than 7 at 25°C ?
 - 10^{-8} M HCl
 - 0.1 M NaHCO_3 ; K_{a1} and K_{a2} for H_2CO_3 are 10^{-6} and 10^{-12} respectively at 25°C
 - 10^{-4} M NaCl
 - 10^{-7} M NaOH
- 100 mL of $\frac{M}{20}$ KMnO_4 can be reduced completely by :
 - 5 millimoles of $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$, in presence of dilute H_2SO_4 .
 - 2.5 millimoles of KI, in presence of dilute NaOH
 - 25 millimoles of HBr in acidic medium
 - 10 millimoles of $\text{Na}_2\text{C}_2\text{O}_4$ in presence of dilute H_2SO_4
- Which of the following is/are pairs of constitutional isomers?
 - 
 - 
 - 
 - 
- Select correct statements out of following:
 - Twist boat form of the cyclohexane is optically active
 - Aniline have five equivalent resonating structures
 - $\text{CH}_3 - \overset{+}{\text{C}}\text{H} - \text{CH} = \text{CH}_2$ and $\overset{+}{\text{C}}\text{H}_2 - \text{CH}_2 - \text{CH} = \text{CH}_2$ are resonating structures
 -  have five equivalent resonating structures

5. Select INCORRECT statements out of following:
- (A) $B(OH)_3$ is a Lewis acid
- (B) Phenolphthalein ($pK_{In} = 9$ at $25^\circ C$) is a suitable indicator to determine end point of the reaction between NH_4OH and HCl
- (C) Basicity of H_3PO_4 is same as that of H_3PO_2
- (D) pH of saturated solution of $Ba(OH)_2$ [$K_{sp} = 4 \times 10^{-6}$] is 12 at $25^\circ C$. [$\log 2 = 0.30$]
6. In which of the following reactions H_2O_2 is acting as an oxidising agent?
- (A) $H_2O_2 + SnO \longrightarrow H_2O + SnO_2$
- (B) $HOCl + H_2O_2 \longrightarrow H_3O^+ + Cl^- + O_2$
- (C) $PbS(s) + 4H_2O_2 \longrightarrow PbSO_4(s) + 4H_2O$
- (D) $H_2O_2 + 2K_2MnO_4 \longrightarrow 2KMnO_4 + 2KOH$

SECTION 2

This section has TWO (02) Paragraphs containing TWO (02) Questions on each paragraph. Each question has FOUR options. ONLY ONE of these four options is the correct answer.

Paragraph for Q-7 to Q-8

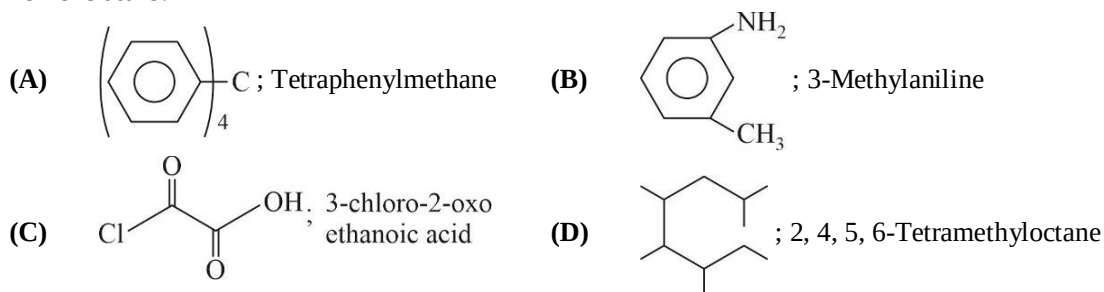
Solubility of a solute in a given solvent can be expressed as moles of the solute dissolved per unit volume of the solution. Usually unit volume of the solution is equal to 1L thus solubility is expressed in term of moles/L. Solubility product (K_{sp}) is the equilibrium constant for dissolution of the solute. For an electrolyte the solubility product is expressed in terms of stoichiometric product of concentration of the ions furnished by the electrolyte. As K_{sp} is an equilibrium constant thus it does not vary by change in active mass of a component. It is not necessary that all of the dissolved part of an electrolyte is fully ionised/dissociated thus care must be taken during theoretical determination of concentration of ions furnished by the weak electrolytes, using their solubilities. Also presence of a strong electrolyte furnishing a common ion can decrease solubility of a weak electrolyte; by common ion effect.

7. $AgBr$ is sparingly soluble in water at $25^\circ C$. If only 80% of the dissolved $AgBr$ get ionised to Ag^+ and Br^- then solubility of $AgBr$ is : (K_{sp} of $AgBr$ at 25° is 10^{-12})
- (A) $1 \times 10^{-6} M$ (B) $1.25 \times 10^{-6} M$ (C) $1.5 \times 10^{-6} M$ (D) $0.8 \times 10^{-6} M$
8. Solubility of $AgBr$ is 1×10^{-y} in presence of $10^{-3} M$ KBr , if presence of the strong electrolyte reduces degree of ionization of $AgBr$ to 10%, then y is equal to : [K_{sp} of $AgBr$ at $25^\circ C$ is 10^{-12}]
- (A) -9 (B) -8 (C) +8 (D) +9

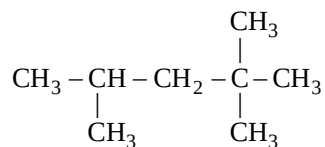
Paragraph for Q-9 to Q-10

According to IUPAC system for the organic compounds, name of an organic compound can be divided into four parts like prefixes, word root, primary suffix and 2° suffix. Prefixes express substituent carbon chains or functional groups; word root gives information of number of C-atoms in the principle carbon chain, primary suffix gives information of saturation/unsaturation of the principle carbon chain; and secondary suffix gives information of principle functional group of the organic compound. To decide the principle group IUPAC had provided a priority order of the functional groups. In case of aromatic compounds some of the common names were retained in the IUPAC system. Numbering of longest principle carbon chain should start with the principle functional group (if present) and should have maximum number of substitutes carbon chains.

9. Which of the following organic compound name is NOT acceptable in the IUPAC system of nomenclature?



10. According to the IUPAC system following compound is named as:



- | | |
|------------------------------|-------------------------|
| (A) Isooctane | (B) Isopropylneopentane |
| (C) 2, 2, 4-Trimethylpentane | (D) All are correct |

SPACE FOR ROUGH WORK

SECTION 3

This section contains 2 Column-Match sets. Each set contains statements given in two columns (I and II). One or more statements in column II can be matched with a statement in column I. Each set has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

11. Match the column :

Column-I (1L aqueous solution)		Column-II (Properties of the solution)	
(A)	1 Litre solution having 0.1 M CH_3COOH ($\text{pK}_a = 5$) and 0.1 M CH_3COONa	(P)	$\text{pH} < 7$ at 25°C
(B)	1 Litre solution having 0.1 M CH_3COONa and 0.1 M HCl (pK_a of $\text{CH}_3\text{COOH} = 5$)	(Q)	$\text{pH} > 7$ at 25°C
(C)	1 Litre solution having 0.1 M CH_3COOH and 0.1 M HCl (K_a of $\text{CH}_3\text{COOH} = 10^{-5}$)	(R)	pH changes with dilution
(D)	1 Litre solution having 0.1 M CH_3COOH and 0.1 M NaOH (K_a of $\text{CH}_3\text{COOH} = 10^{-5}$)	(S)	pH do not change with dilution

(A) [A-P, S] [B-P, R] [C-Q, R] [D-Q, S]

(B) [A-P, R] [B-Q, R] [C-Q, S] [D-P, S]

(C) [A-P, S] [B-P, R] [C-P, R] [D-Q, R]

(D) [A-Q, S] [B-P, S] [C-P, R] [D-Q, R]

12. Match the column:

Column-I (Oxidation or Reducing species)		Column-II (Properties of the species)	
(A)	MnO_4^- in acidic medium	(P)	Equivalent mass of the species as reducing agent is equal to $\frac{M^\circ}{2}$ (M° is molar mass of that species)
(B)	$\text{Cr}_2\text{O}_7^{2-}$ in acidic medium	(Q)	Equivalent mass of the species as oxidising agent is less than $\frac{M^\circ}{2}$ (M° is molar mass of that species)
(C)	$\text{C}_2\text{O}_4^{2-}$ in acidic medium	(R)	Element (other than oxygen) in the species is in its highest oxidation state
(D)	H_2O_2 in acidic medium	(S)	Species having a peroxide bond

(A) [A-Q, S] [B-P, R] [C-P, R] [D-Q, S]

(B) [A-P, R] [B-Q, R] [C-P] [D-P, S]

(C) [A-P, R] [B-R, S] [C-P, R] [D-Q, S]

(D) [A-Q, R] [B-Q, R] [C-P] [D-P, R, S]

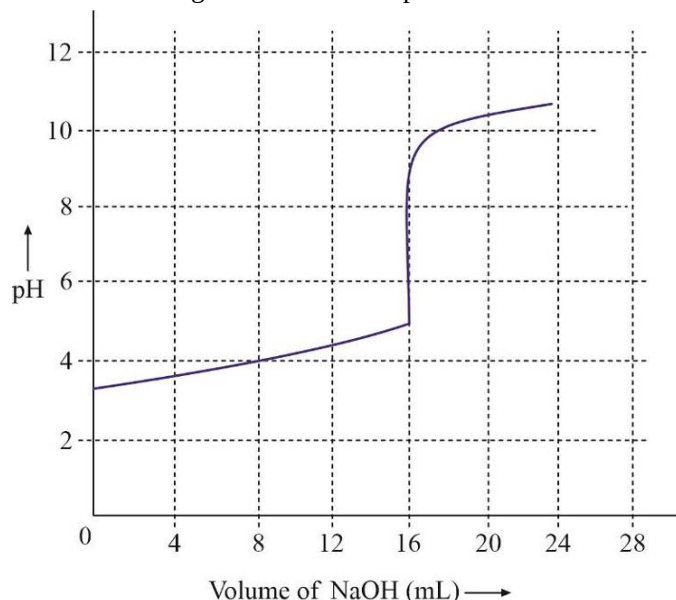
SPACE FOR ROUGH WORK

SECTION 4

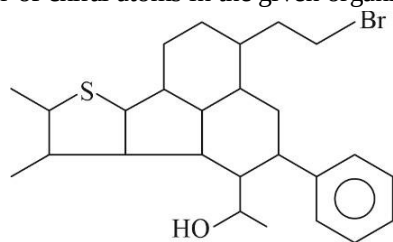
This section has Six (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.

(Example: 6, 81, 1.50, 3.25, 0.08)

13. Total number of isomers of C_5H_{10} (including stereoisomers) is _____.
14. A weak organic acid (HX) is titrated with NaOH and variation of pH of the solution with addition of NaOH is shown in the figure. What will be pH of 1M NaX solution at 25°C ?



15. 50 mL of $\frac{M}{10}$ $Na_2S_2O_3$ is sufficient to decolourise a given starch iodine solution. After the reaction, the solution is carefully treated with ozone. Volume (in mL) of the ozone (at STP) that would be required to completely convert iodide content of the solution into iodate is _____.
(At STP, molar volume of a gas is 22.4 L)
16. On treatment with H_2O_2 in an alkaline medium, $K_2Cr_2O_7$ got converted into $K_3[Cr(O_2)_4]$. An anion of this compound consist of four peroxide bonds. During the reaction H_2O_2 got oxidised to $O_2(g)$. Moles of H_2O_2 required to completely reduce 1 mole of $K_2Cr_2O_7$ in this reaction is.
17. 100 mL of 1 M H_3PO_3 solution was titrated with 1 M KOH solution.
If pH at 1st equivalent point is x and that of end point is y then value of $(y - x)$ is _____.
[Given that for H_3PO_3 at 25°C pK_{a_1} and pK_{a_2} are 2.5 and 6.5 respectively, $\log 2 = 0.30$, $\log 3 = 0.50$]
18. Number of chiral atoms in the given organic compound is _____.



PART III : MATHEMATICS**60 MARKS****SECTION 1**

This section consists of SIX (06) Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

- 8 Arithmetic means are inserted between 12 and 75. Which of the following is(are) correct?
 - Arithmetic mean of the 8 arithmetic means is 42.5
 - 4th arithmetic mean is 40
 - Second last arithmetic mean is 63
 - The sum of second and fourth arithmetic means is 66
- Consider the statements about $y = \frac{\sin 3x}{\sin x}$ ($\sin x \neq 0$). Which of the following is(are) correct?
 - The minimum value of y is (-1)
 - The maximum value of y is 3
 - The minimum value of y is NOT defined
 - The maximum value of y is NOT defined
- Choose the correct option(s):
 - The value of $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7}$ is $\left(-\frac{1}{8}\right)$
 - The value of $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7}$ is $\left(-\frac{1}{2}\right)$
 - The value of $\cos \frac{2\pi}{7} \cos \frac{4\pi}{7} \cos \frac{6\pi}{7}$ is $\left(\frac{1}{8}\right)$
 - The value of $\cos \frac{2\pi}{7} \cos \frac{4\pi}{7} \cos \frac{6\pi}{7}$ is $\left(\frac{1}{2}\right)$
- Consider the ellipse having equation $\frac{x^2}{9} + \frac{y^2}{4} = 1$. Choose the correct option(s):
 - The axes are parallel to coordinate axis
 - The length of latus rectum is less than 1
 - The distance between foci is irrational
 - The distance between directrices is less than 8
- The common tangents to the circle $x^2 + y^2 = 2$ and the parabola $y^2 = 8x$ touch the circle at the points P, Q and the parabola at the points R, S. Which of the following statement(s) is true? (P & S are above x -axis)

(A) The length of PS is $3\sqrt{2}$ units	(B) Area of trapezium is 15 units
(C) The length of QS is $\sqrt{26}$ units	(D) Area of triangle PQR is 3 units
- Let a, b, c be positive integers such that $\frac{b}{a}$ is an integer. If a, b, c are in geometric progression and the arithmetic mean of a, b, c is $(b+2)$, then choose the correct option(s):

(A) The value of a is a composite number	(B) The value of a is a prime number
(C) The value of $\left(\frac{b}{a}\right)$ is a prime number	(D) The value of $\left(\frac{b}{a}\right)$ is a composite number

SPACE FOR ROUGH WORK

SECTION 2

This section has **TWO (02) Paragraphs** containing **TWO (02) Questions** on each paragraph. Each question has **FOUR options**. **ONLY ONE** of these four options is the correct answer.

Paragraph for Q-7 to Q-8

$y^2 = 4x$ and $y^2 = -8(x - a)$ intersect at point A and C. Points $O(0, 0)$, A, $B(a, 0)$ and C are concyclic ($a > 0$).

7. The length of the common chord of parabolas is:

- (A) $2\sqrt{6}$ (B) $4\sqrt{3}$ (C) $6\sqrt{5}$ (D) $8\sqrt{2}$

8. The area of cyclic quadrilateral $OABC$ is:

- (A) $24\sqrt{3}$ (B) $48\sqrt{2}$ (C) $12\sqrt{6}$ (D) $18\sqrt{5}$

Paragraph for Q-9 to Q-10

Box 1 contains three cards bearing numbers 1, 2, 3 ; box 2 contains five cards bearing numbers 1, 2, 3, 4, 5 ; and box 3 contains seven cards bearing numbers 1, 2, 3, 4, 5, 6, 7. A card is drawn from each of the boxes. Let x_i be the number on the card drawn from i^{th} box, $i = 1, 2, 3$.

9. The number of draws such that $x_1 + x_2 + x_3$ is odd is:

- (A) 29 (B) 25 (C) 53 (D) None of these

10. The number of draws such that x_1, x_2, x_3 are in arithmetic progression is:

- (A) 7 (B) 8 (C) 10 (D) 11

SPACE FOR ROUGH WORK

SECTION 3

This section contains 2 Column-Match sets. Each set contains statements given in two columns (I and II). One or more statements in column II can be matched with a statement in column I. Each set has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

11. Match column I to Column II according to the given condition

	Column I		Column II
I.	Tangents at the ends of focal chord of a parabola	(P)	Meet at directrix
II.	Let P is a point on the parabola and tangent at P meets the directrix at Q. If S is the focus of parabola then SP and SQ	(Q)	Meet at an angle of 90°
III.	If $A(at_1^2, 2at_1)$ and $B(at_2^2, 2at_2)$ are the ends of a focal chord of $y^2 = 4ax$, then	(R)	$t_1 t_2 = -1$
IV.	If the normal at $(at_1^2, 2at_1)$ meets the parabola $y^2 = 4ax$ again at $(at_2^2, 2at_2)$, then	(S)	$t_2 = -t_1 - \frac{2}{t_1}$
		(T)	$t_1 t_2 + t_1^2 + 2 = 0$

(A) [I - P, Q] [II - Q] [III - R] [IV - S, T] (B) [I - P, S] [II - Q] [III - R] [IV - S, T]

(C) [I - P, Q] [II - R] [III - S] [IV - S, T] (D) [I - P, R] [II - S] [III - Q] [IV - S, T]

12. MATCH THE FOLLOWING COLUMNS:

	Column I		Column II
I.	$1 \cdot 2 \cdot 3 \cdot 4 + 2 \cdot 3 \cdot 4 \cdot 5 + \dots + n(n+1)(n+2)(n+3) + \dots$ to 11 terms	(P)	10385
II.	$\frac{1^3}{1} + \frac{1^3 + 2^3}{1+3} + \dots + \frac{1^3 + 2^3 + \dots + n^3}{1+3+\dots+(2n-1)} + \dots$ to 16 terms	(Q)	37820
III.	$1 \cdot 3 + 2 \cdot 4 + \dots + n(n+2) + \dots$ to 30 terms	(R)	446
IV.	$1 + 3 + \dots + \frac{n(n+1)}{2} + \dots$ to 60 terms	(S)	72072

(A) [I - P] [II - Q] [III - S] [IV - R] (B) [I - S] [II - P] [III - R] [IV - Q]

(C) [I - S] [II - R] [III - P] [IV - Q] (D) [I - S] [II - Q] [III - P] [IV - R]

SPACE FOR ROUGH WORK

SECTION 4

This section has Six (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.*

(Example: 6, 81, 1.50, 3.25, 0.08)

13. Coefficient of x^{11} in $(1+x^2)^4(1+x^3)^7(1+x^4)^{12}$ is n . The product of digits in n is ____.
14. For $1 \leq m \leq n$, if $f(m, n) = {}^nC_0 - {}^nC_1 + {}^nC_2 - \dots + (-1)^{m-1} \cdot {}^nC_{m-1}$, then $f(7, 8)$ is ____.
15. Let $n \geq 2$ be an integer. Take n distinct points on a circle and join each pair of points by a line segment. Colour the line segment joining every pair of adjacent points by blue and the rest by red. If the number of red and blue line segments are equal, then the sum of digits in $n!$ is ____.
16. The number of distinct solutions of the equation $\frac{5}{4}\cos^2 2x + \cos^4 x + \sin^4 x + \sin^6 x + \cos^6 x = 2$ in the interval $[0, \pi]$ is ____.
17. The sum of first 9 terms of the series $\frac{1}{\sqrt{1}} + \frac{1+2}{\sqrt{1+3}} + \frac{1+2+3}{\sqrt{1+3+5}} + \dots$ is n . The unit digit in n is ____.
18. The minimum value of the sum of real numbers $a^{-4}, a^{-3}, 4a^{-2}, 1, a^7, a^8$ with $a > 0$ is ____.

SPACE FOR ROUGH WORK

End of JEE Advanced Booster Test - 4 | 2022